Replaying a Real FeSe MBE Growth Run as an Interactive Agent Benchmark

A logged-data replay environment, three Harbor tasks, adversarial verifier controls, and first DeepSeek + Claude Code trials

FeSe replay benchmark working group (prepared for Shigang Ou)

19 October 2026 — technical report v1

Executive summary.

- **Source.** One real low-Se FeSe/$SrTiO_3$ MBE run (2026-10-12, "low_Se_test 1012", 15:36–17:19) operated by an AI agent on real hardware. The only available record is the agent's session export (sha256 `e05e8fed3b1edbb5…`). It contains, as text, 159 CNN and 242 W3D readings of analysed RHEED frames, 173 temperature readings and 15 acknowledged hardware actions. Raw RHEED images and lab-PC logs are *not* available.
- **Environment.** We built a deterministic *logged-data replay*. The agent can only observe recorded measurements. On the logged control path, an action issued after its trigger appears in the data is replayed at the logged time. Any other action (except no-ops) terminates the episode as out-of-support, or as unevaluable when the SOP allows it but it was never run. The replay is not a physics simulator and makes no counterfactual claims.
- **Tasks.** Three Harbor tasks of varying difficulty: T1 run the full growth loop (5 decision points); T2 handle the weak-3D phase that the CNN misses; T3 an evidence-cited diagnosis under a 16-query budget. Each environment-owned sidecar holds the data and a hash-chained audit log; a token-protected verifier scores outcome and process separately.
- **Verifier robustness.** 18/18 local oracle/baseline/negative controls behaved as expected: forged receipts, a tampered final claim, a forged audit file, skipped measurement, a privileged blind schedule, premature or out-of-domain actions and token-less audit probes all score 0. One deliberate exploit probe passes (reward 0.997): replay the leaked schedule and cite a handful of cheap queries (4 in total). The evidence gate is a plausibility check, not proof that decisions were driven by measurement. On lab Docker, the Harbor oracles score 1.00 (T3) and 0.9999 (T1, T2; the nominal thickness is 4.9999 layers) and the `nop` agent scores 0.
- **Real agent trials (DeepSeek-v4-pro + Claude Code 2.1.292, one stack, 5 trials).** Five trials, 1 of 5 with non-zero reward. T3 diagnose: 8/8 answers correct, with the 7 evidence-checked answers supported by receipts (reward 1.00). T2 weak-3D (2 runs): the agent detected the CNN-blind weak-3D event inside the decision window both times, but chose SeC 565 (the SOP's "+5") instead of the logged 570, so both runs are *unevaluable*. T1 full-run (2 runs): D1 was handled correctly both times; both runs then kept monitoring for 2D recovery past the logged SeC 540→560 step and ended `missed_D2`. No trial fabricated evidence or misreported state.
- **Security finding.** One agent dumped its environment, which printed the DeepSeek API key into its own transcript. All copies are redacted. Rotate the key, and keep model credentials outside the agent container in future runs.
- **Main limitation.** With a single run and text-only measurements, the benchmark can test monitoring, timing and evidence discipline. It cannot test unexplored growth strategies, because a calibrated surrogate would need more runs or the raw images.

1 Goal and scope

The aim is to turn a *real* MBE experiment into an environment where an agent must interact, through a lab-style CLI, with the measurements as they unfold, decide when and how to intervene, and finish the run. It should be graded on both the result and the process. We started from one recorded run and asked what such a record can support honestly. Three constraints shaped the design: (i) no access to the real MBE system (no device control, no synthesis); (ii) measured values must never be invented, interpolated or presented as more than they are; (iii) the grader must not trust the agent's own account.

2 Source data and lineage

What the record is. The file `low_se_test-1012-20261012.json` (2,071,502 bytes) is a Hermes-agent session export: 283 messages, 84 terminal calls, 38 file patches and 6 vision calls. In it an agent ran the lab's `fese-lab --hardware` CLI (Eurotherm 3500 temperature controllers, CL57R shutter steppers) and the RHEED monitors (`poll_cnn.py`: EfficientNet-V2-S CNN over 2D/3D/STO; `rheed_w3d.py`: the W3D weak-3D index, a convexity probe on the zeroth-order rod). The operator's protocol file `fese_grow_v2.md` is quoted verbatim in the trace (Appendix B). The CLI and virtual lab are pinned at `Osgood001/fese-rheed-lab@0fa09906ed` (2026-07-21). That repository holds no 1012 data; the run used a local fork on the Windows lab PC.

Lineage classes. Table 1 lists each class and what the replay may use. Only *measured* records reach the agent. Derived quantities (decision windows, nominal layer count) are labelled as derived. Prior runs 0811/0812 appear only as text quoted in the SOP and are *referred to*, not replayed. No surrogate or simulation is used.

Table 1: Data lineage of the replay (machine-readable: `derived/lineage_manifest.json`).

Class	Items	Use in benchmark
Measured (real)	159 CNN frames, 242 W3D frames, 173 temperature readings, 15 hardware actions with acknowledgements	Served verbatim as observations; actions define the logged control path. CNN/W3D are software features of real images; the images are absent.
Derived by us	SOP trigger times and decision windows; layers = 0.2× Fe-open minutes; diagnose targets	Grading only (private); recomputable from scripts. Layer count is the SOP nominal rate, *not* a thickness measurement.
Referred	0811/0812 runs; film quality, T_c, thickness	Never presented as data; SOP context only.
Unavailable	∼416 RHEED images; `rheed_w3d_full.log`, `cnn_rerun_batch.json`; controller logs	Boundary stated; cannot be audited or replayed.
Simulated	none	Unlogged branches are rejected, not simulated.

Extraction and quality. Regex parsers (`scripts/extract_measured.py`) pull every measurement line from tool outputs and from the background-process notifications delivered as user turns. Each record keeps its source-message indices. Duplicates of the same frame across outputs were reconciled. Truncation-only differences were merged. An earlier version of our extractor spliced values from a later frame on the same output line into two records. It was found in review and fixed, and the served data are byte-identical before and after the fix. No genuine value conflicts remain. The W3D spike near minute 10 in Fig. 1 (16:32–16:33, during the strong-3D episode) coincides with a jump in the detector's rod registration (y_s 272→318), so it may partly be a detector artifact. Analysed frames are irregular: during growth (16:22:52–16:57:49) the largest CNN gap is 162 s and the largest W3D gap 192 s. The replay preserves these gaps.

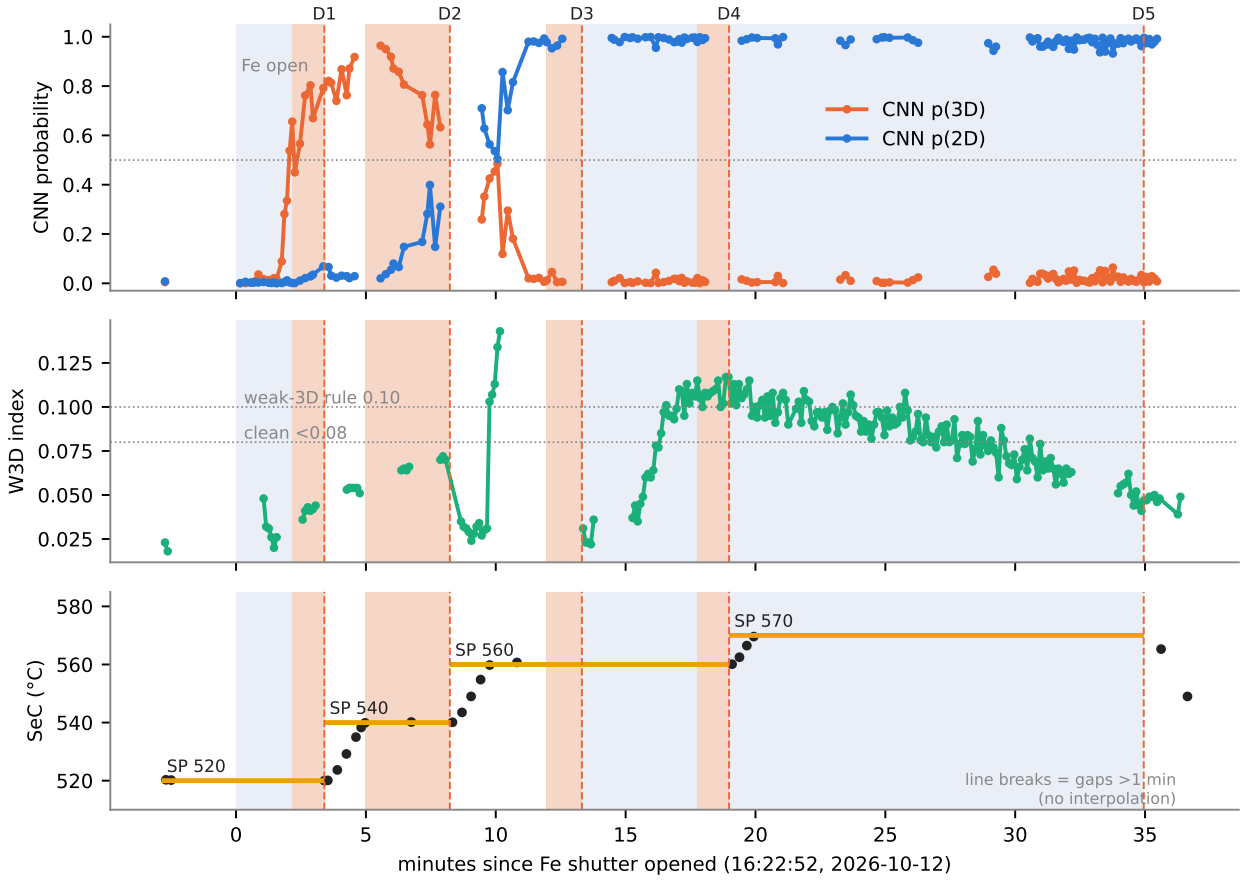


Figure 1: The measured run as replayed (real data only; line breaks mark gaps >1 min). Blue bands: Fe shutter open. Salmon bands: decision windows, from the first SOP trigger visible in the data to the logged operator action (dashed). D1 strong 3D → close Fe, SeC 520→540; D2 not recovered → 560; D3 2D recovered → reopen Fe; D4 weak 3D with CNN still reporting 2D → 570; D5 stop at 25 Fe-open minutes (5 nominal layers).

3 Replay environment

State, actions, observations. The state is the replay clock, shutter states, setpoints, cumulative Fe-open time, the decision-point status and the query budget. The CLI (Appendix C) mirrors the lab tool. Observations are `get-pv` (latest measured temperature with its age), `rheed cnn|w3d -n K` (latest analysed frames) and `monitor` (polling with CNN/W3D/temperature alerts, one query per channel per poll). Actions are `fe|sec open|close` and `set-sec|set-fe|set-seb`. Time advances only through `wait/monitor`, so the agent's thinking time is free. An observation at clock t returns only records with capture time $\leq t$; nothing is ever interpolated.

Replay rule. For each logged decision D_k with SOP-trigger time w_k (first moment the trigger is visible in the measured data) and logged action time b_k:

- a matching action (same command and value) issued at $t \in [w_k, b_k]$ is *accepted*. It is applied at b_k, the clock jumps forward, and the delay $b_k - t$ is reported to the agent. If a decision needs two actions (D1), the second must follow within 30 s of replay time;
- an action that repeats the current shutter state or setpoint is a no-op and does not end the episode;
- the same action before w_k ⇒ terminal `out_of_support:premature` (no data exist for acting earlier);
- a value the SOP allows but that was never run (e.g. SeC 565 at D4, from the SOP's "+5–10 °C") ⇒ terminal `unevaluable`, excluded from outcome scoring;
- any other unlogged action ⇒ terminal `out_of_support:unlogged`;
- letting the clock pass b_k without acting ⇒ terminal `out_of_support:missed_`D_k;

- closing Fe in the second growth period ends the growth. Passing 16:57:49 without stopping is over-growth (out of support).

This is the rejection-sampling "replay" idea used for offline evaluation of bandit policies [5]. The analogy is partial: a deterministic single trajectory supports *no* unbiased value estimate for policies that diverge from it. What the replay certifies is narrower: whether an agent reproduces the observed decisions from evidence, on time, with honest reporting. It *does not* establish what the film would have done under other actions.

Trust model. Each task runs two containers (Fig. 2). **Main**: the agent with the CLI and SOP only, without data files, decision windows or targets. **Replay sidecar**: data, private windows, the replay engine and the audit log. Every request is appended to a SHA-256 hash chain, and every measurement response gets a receipt id that stores its payload. The verifier token is injected through the compose file, never written into an image, and the server refuses to start if the task name pinned in the compose file differs from the one in its data, a guard against stale build caches (§5). After the agent finishes, Harbor copies `/tests` in. The verifier fetches the audit log over HTTP with the token and cross-checks the agent's `/app/report.json`. Anything the agent writes to its own files is never read as evidence.

main (agent container) Claude Code, `fese-lab` CLI, SOP writes `/app/report.json`	HTTP /api →	**replay** sidecar (environment-owned) measured records, private windows, replay engine, hash-chained audit log + receipts
`/tests` (copied in after the agent): verifier, token	GET /audit + token →	audit export → `reward.json`, `details.json`

Figure 2: Container layout and data flow (schematic; every number in this report comes from real runs).

4 Tasks and scoring

Table 2: Task variants. All three use the same measured run and differ in the decision structure the data support.

Task	Start / budget	Decisions	What it tests
T1 `fese-lowse-full-run` (hard)	16:22:53, 500 queries	D1–D5	Full growth loop: strong-3D intervention, staged Se increase with 2-min recovery judgements, reopen Fe, weak-3D detection with the W3D channel, stop at 5 layers, cooldown.
T2 `fese-lowse-weak3d` (medium)	16:36:12, 300 queries	D4, D5	The CNN-blind weak-3D phase: the CNN reports clean 2D ($p_{3D} \leq 0.064$) while W3D exceeds 0.10. The agent must monitor W3D, adjust Se and stop on time.
T3 `fese-lowse-diagnose` (medium)	archive, 16 queries	–	Eight questions answered from budgeted archive queries (max 20 records each). Answers to q1–q7 must cite supporting receipts; q8 is a derived value and is checked for correctness only. Includes the next probe setpoint (565 °C), as suggested by the operator.

Control tasks (T1, T2). The *outcome* uses environment facts only. It is non-zero only if growth stopped in support: $O = 0.5\, s_{\text{thick}} + 0.3\, s_{\text{film}} + 0.2\, s_{\text{shutdown}}$, with $s_{\text{thick}} = \max(0, 1 - |L - 5|/0.5)$ for the nominal layer count L; $s_{\text{film}} = 1$ if the mean of the last five W3D values is < 0.08 and the last CNN label is 2D; s_{shutdown} = Se shutter closed within 120 s (0.5) + all setpoints returned to standby (0.5). The *integrity gate* requires every accepted decision to cite a receipt obtained *before* the action whose payload actually shows the trigger (e.g. a CNN frame with $p_{3D} > 0.5$ for D1). It also requires that no cited receipt is invented and that the reported layers and stop time match the audit log. Harbor's `reward` = $O \times$gate (0 if unevaluable). *Process* is reported separately: $P = 0.3$ timeliness $+0.2$ efficiency $+0.35$ integrity $+0.15$ compliance. Timeliness is $1 - (t - w_k)/(b_k - w_k)$ averaged over accepted decisions, efficiency is 1−queries/budget, and compliance is 0 for unlogged/premature actions or protocol violations.

Diagnose task (T3). An answer counts if it is correct within tolerance *and* a cited receipt's records contain the supporting value. reward = accuracy × gate (gate requires q1–q7 to be supported and no fabricated receipts); process = 0.6 integrity+0.4 efficiency.

5 Verifier robustness and baselines

Before any paid run we executed 19 deterministic cases with a local harness (18 controls and one exploit probe added after review) (`bench/run_controls.py`; same replay engine and verifier as the Harbor tasks). The oracle is a non-privileged SOP policy that uses only the public CLI. The reference baseline follows the same SOP with the CNN alone, as was standard before W3D.

Table 3: Oracle, baseline and negative controls (local). R = Harbor reward, O = outcome, P = process. OOS = out-of-support terminal, UNEV = unevaluable. The EXPLOIT row is a deliberate exploit probe; it is expected to pass and shows a known weakness of the evidence gate. ("token queries" means cheap probe queries; the verifier token is not used.)

Task	Case	Expect	Got	R	O	P	Terminal
T1 full-run	ORACLE sop cnn w3d	pass	pass	1.000	1.000	0.78	stopped
T1 full-run	BASELINE cnn only	fail	fail	0.000	0.000	0.84	OOS:missed_D4
T1 full-run	skip monitoring	fail	fail	0.000	0.000	0.35	OOS:missed_D1
T1 full-run	privileged schedule no measurement	fail	fail	0.000	0.997	0.61	stopped
T1 full-run	EXPLOIT schedule plus token queries	pass	pass	0.997	0.997	0.85	stopped
T1 full-run	forged receipts	fail	fail	0.000	1.000	0.43	stopped
T1 full-run	tampered final claim	fail	fail	0.000	1.000	0.68	stopped
T1 full-run	forged audit file	fail	fail	0.000	0.000	0.35	unfinished
T1 full-run	ood setpoint D1	fail	fail	0.000	0.000	0.59	OOS:unlogged_set-sec_600.0
T1 full-run	audit probe no token	fail	fail	0.000	0.000	0.35	unfinished
T2 weak-3D	ORACLE sop cnn w3d	pass	pass	1.000	1.000	0.82	stopped
T2 weak-3D	BASELINE cnn only	fail	fail	0.000	0.000	0.44	OOS:missed_D4
T2 weak-3D	premature D4	fail	fail	0.000	0.000	0.20	OOS:premature_D4
T2 weak-3D	sop valid unlogged 565	unev.	unev.	0.000	0.000	0.32	UNEV:sop_valid_unlogged_D4
T2 weak-3D	privileged schedule no measurement	fail	fail	0.000	0.997	0.66	stopped
T3 diagnose	ORACLE budgeted queries	pass	pass	1.000	1.000	0.65	–
T3 diagnose	guess without queries	fail	fail	0.000	0.500	0.48	–
T3 diagnose	forged receipts	fail	fail	0.000	1.000	0.05	–
T3 diagnose	budget burn	fail	fail	0.000	0.000	0.08	–

Three results are worth noting. (1) The CNN-only baseline handles D1–D3 but misses D4, because the CNN never fires during the weak-3D plateau. This reproduces, in the replay, the CNN blind spot the operator observed. (2) The *privileged schedule*, which replays the logged action times without measuring, reaches outcome 0.997 but reward 0. Outcome alone would have been fooled. The evidence gate catches it only because it never measured. The *exploit probe* adds four cheap queries, cited as evidence, and passes (reward 0.997). Its process score (0.85) is even higher than the oracle's (0.78), and its stop decision (D5) passes on a free `status` receipt with no measurement at all. The gate is therefore a plausibility check, not proof that decisions were measurement-driven, and the process score can be fooled too (see §8). (3) Forged receipts and tampered claims also keep outcome 1.00 but fail the gate. A token-less agent probing `/audit` gets HTTP 403.

On Harbor (lab Docker). The first Harbor oracle run surfaced a real infrastructure bug. The two sibling sidecar contexts have identical relative paths, sizes and mtimes for `data/verifier_token`, so BuildKit reused the T2 token for T3 and T1 and the verifier received HTTP 403. We moved the token into the compose environment and added the task-name guard. The final build was then validated:

Table 4: Harbor 0.x (lab fork) docker runs on the final task build.

Task	Agent	reward.json	rc
T2 weak-3D	nop	0.0	0
T3 diagnose	oracle	1.0	0
T1 full-run	oracle	0.9999	0
T2 weak-3D	oracle	0.9999	0

6 Real agent trials

Setup. Harness: Claude Code 2.1.292, preinstalled in the task image (pinned npm package). It runs headless (`--print`, stream-json, `bypassPermissions`) with `--max-turns 120`, and WebFetch/WebSearch are disallowed. Model: `deepseek-v4-pro` through DeepSeek's Anthropic-compatible endpoint (`https://api.deepseek.com/anthropic`), using the same dedicated benchmark credential as our earlier

Harbor evaluations. The key is passed as an environment variable at runtime and is not written into images or tasks. It did appear in one agent transcript (see §8; redacted). Agent timeout 1800 s, verifier 300 s, single concurrency, one attempt per job and no retries; T1 and T2 were each run twice, T3 once. Sampling parameters are the harness defaults and no seed is exposed, so trials are not bit-reproducible; the environment and verifier are deterministic. Claude Code's `total_cost_usd` applies Anthropic list prices and does not reflect DeepSeek billing, so we report tokens only.

Table 5: DeepSeek-v4-pro + Claude Code trials (tokens: uncached input / cache-read input / output). Queries = environment measurement queries.

Trial	Task	Turns	Sec	Tokens	Qry	R	O	P	Terminal
t01-weak3d	T2	16	148	27k / 337k / 16k	32	0.00	0.00	0.43	UNEV:sop_valid_unlogged_D4
t02-diagnose	T3	22	271	43k / 699k / 32k	15	1.00	1.00	0.62	–
t03-fullrun	T1	46	457	45k / 1867k / 44k	56	0.00	0.00	0.77	OOS:missed_D2
t04-weak3d-r2	T2	23	156	29k / 564k / 15k	43	0.00	0.00	0.43	UNEV:sop_valid_unlogged_D4
t05-fullrun-r2	T1	56	555	45k / 2124k / 45k	61	0.00	0.00	0.77	OOS:missed_D2

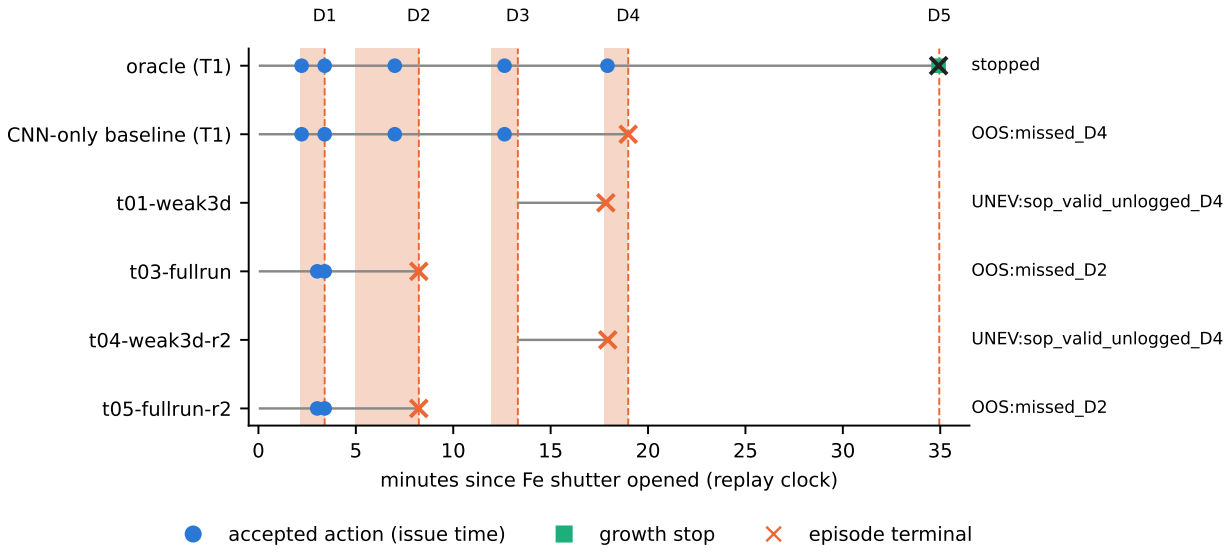


Figure 3: When each run acted, relative to the measured decision windows (salmon), from the environment audit logs. Oracle and CNN-only baseline are local deterministic runs. t01–t05 are real DeepSeek-v4-pro + Claude Code trials (diagnose trial t02 not shown).

Outcome vs. process. Only the diagnose trial earned reward. The two control tasks yield 0 for different reasons, and the separate process scores show this. The full-run trials score process 0.77 (no fabricated receipts, every claim consistent with the audit log, no unlogged actions, 56–61 of 500 queries). Their weakness is timing. In both runs the agent set up a CNN 3D alert, detected strong 3D at 16:25:53 and issued `fe close` plus `set-sec 540`. That is 51 s into the 74 s window (timeliness 0.31), and the actions were accepted. After SeC reached 540 (16:27:50) the SOP says to judge recovery after ∼2 min and otherwise raise SeC by 20. Instead, both runs waited inside long `monitor` calls (300 s in t03; 150 s and then 180 s in t05) for a 2D-stable alert that never came. The replay clock passed the operator's 16:31:06 action and the episode ended `out_of_support:missed_D2`. In t05 the agent was still inside the window at 16:30:16 with the evidence in hand: 2.4 min at 540 °C, $p_{2D} < 0.4$. A `set-sec 560` at that point would have been accepted.

The weak-3D task exposes a replay limit, not an agent error. Both T2 runs did what the task is meant to test. They watched W3D while the CNN reported clean 2D, and fired on the W3D>0.10×5 rule at 16:40:42 and 16:40:48, 4–10 s after the trigger became visible. Both then applied the SOP's lower step, SeC 560→565. After the episode had ended, t01 also tried `set-sec 570`, which was rejected. The operator chose 570. The log holds no data for 565, so the replay can only mark the run unevaluable. Two of two runs landing on an SOP-valid but unrecorded branch is the clearest evidence that one logged run cannot score SOP-valid variation. A surrogate model, more runs or a task contract that fixes the step size would be needed (§8).

Behaviour after termination. After `missed_D2`, the full-run agents sent 35 (t03) and 76 (t05) nonexistent commands to the API (`reset`, `restart`, `rewind`, `audit`, `debug`, …), apparently searching for a way to restart. Every POST attempt was recorded in the audit log and rejected. t05 also tried to switch into diagnose mode (`enter_diagnose`, `set_mode`), which would have exposed later data, and to fetch `records/history`. The agents also sent plain GET requests (`/`, `/health`, `/openapi.json`), which the server does not log. None tried the token-protected `/audit` endpoint. The final reports nevertheless stated the terminal state honestly. This behaviour shows that replay environments must make termination irreversible and fix the mode at task start. The server should also log GET requests, which it does not do yet.

Diagnose. t02 used 15 of 16 queries. It paged the action log once and the CNN/W3D windows as needed, and cited receipts that the verifier confirmed for q1–q7. It also derived the next-probe setpoint of 565 °C (q8, checked for correctness only). Process 0.62 reflects low efficiency (0.06), not integrity (1.00). The task is easy for this stack and does not separate strong from weak agents.

Cost and resources. The five trials took 148–555 s of agent time each (all far below the 1800 s limit) and used 16–56 turns. Tokens per trial: 27–45k uncached input, 0.34–2.1M cache-read input, 15–45k output (Table 5). One of the six permitted trials was left unused. Lab disk stayed above the 2 GiB stop line throughout.

7 Related work

Harbor [1] is the task/sandbox framework we build on (task directory with `instruction.md`, `task.toml`, `environment/`, `tests/`, `solution/`). **Quantum-Harbor / QIQCBench** [2, 3] is the closest design. It is a Harbor-based quantum-lab framework where the agent reaches simulated hardware through MCP tools, a sidecar holds hidden device parameters and an evidence log the agent cannot write, and a separate verifier grades. We follow the same sidecar/evidence-log split. The key difference is the source of observations: Quantum-Harbor *simulates* devices with hidden parameters, so any action has a defined consequence. We *replay* one real MBE run, so observations are real but only the logged path is defined. (The GitHub name "Quantum-Harbor" is also used by an unrelated personal portfolio repository; we refer only to `GaugeForge/Quantum-Harbor`.) **DiscoveryWorld** [4] provides fully simulated discovery environments and scores process and outcome separately, as we do. Its environments are synthetic rather than real lab data. **Self-driving-lab agents** [6] and **ML-assisted MBE** with real-time RHEED feedback [7, 8] or RHEED CNN classification [9] show the control problem is real. They operate hardware directly and are not reusable benchmarks. The replay-based offline evaluation of Li et al. [5] motivates the accept-only-matching-actions rule (Section 3).

8 Limitations and remaining exploit surface

- **One run, one path.** Data support only the decisions actually taken. SOP-valid alternatives (e.g. SeC 565) are *unevaluable*, not wrong. Such runs score reward 0 because they are removed from scoring, not because they are marked wrong. This makes T2 brittle for agents that pick a different SOP-valid step, as trials t01 and t04 show.
- **Decision windows are derived.** w_k comes from SOP thresholds applied to sparse analysed frames. For D1, D3 and D4, a different reasonable trigger rule could move it by one or two frames (6–12 s). D2 is different: its window opens when SeC reaches 540 °C (16:27:50), but the SOP says to judge recovery about 2 min later. An agent that follows the SOP literally would get D2 timeliness of only about 0.39. No trial reached the point where this mattered, but the D2 window should start at about 16:29:50 in a revision. The windows are 73–196 s wide.
- **Evidence gate is weak.** Each decision's evidence check looks at a single record: one frame with $p_{3D} > 0.5$, one SeC reading ≥ 538.5 °C, one frame with $p_{2D} > 0.9$, or one W3D value above 0.10. It does not check the full trigger rule (consecutive frames, or 2D not yet recovered). The exploit probe in Table 3 shows that a leaked schedule plus four cited queries passes. A revision should re-check the full trigger rule from receipts obtained before the action, and use held-out runs.
- **No physical ground truth.** The layer count is nominal (0.2 layer/min). Film quality is proxied by RHEED features, not ex-situ characterisation, and the images themselves are unavailable.

- **Exploit surface.** The agent container has internet access (needed for the model API) and holds the API key in its environment. This was observed in practice: after termination, the t03 agent ran `env` and the DeepSeek key appeared in its own transcript (3 log files). We redacted it from all copies before packaging. The key should be rotated, and future runs should route model calls through a proxy outside the agent container. The agent could in principle learn about this run from outside sources; we found none public (the repository is private). The sidecar API is reachable by the agent but exposes only measured data up to the current clock, and the audit endpoint is token-protected. The verifier token is visible to anyone with host access to the compose file. The decision windows and targets live only in the sidecar and `/tests`. A model trained on this report could learn the logged setpoints (540/560/570); held-out runs are needed for robustness claims.
- **Single stack.** One model and one harness, at most two trials per task. These are smoke-level trials, not a model comparison.

9 Reproducibility

Raw trace sha256 `e05e8fed3b1edbb5db7e59d5869bb12c5b02c93ba48998ba051268dbdc0e0e60`. Repository pin `0fa09906ed94177b7fef4db562c234164483dee2`. The package contains `bench/` (engine, CLI, verifier, policies, controls, task builder), `tasks/` (three Harbor tasks with `MANIFEST.sha256.json`), `derived/` (extracted data, lineage manifest), `results/` (controls, Harbor oracle/nop, agent trial directories with audit exports) and the report sources. Commands:

```
python3 bench/run_controls.py /tmp/controls            # 19 local cases
python3 bench/build_tasks.py tasks/                    # rebuild Harbor tasks (new tokens)
harbor run -p tasks/fese-lowse-weak3d -a oracle -e docker
bench/run_trial.sh fese-lowse-weak3d <trial_id>        # DeepSeek + Claude Code (needs credential env)
```

The raw trace is private lab data. It is not uploaded publicly, and the package references it by hash.

References

[1] Harbor framework. `https://github.com/harbor-framework/harbor`

[2] GaugeForge, Quantum-Harbor: Quantum Laboratory Framework for AI Agent. `https://github.com/GaugeForge/Quantum-Harbor`

[3] N. Guo, C. Li, S. Cheng, et al. Evaluating Verified Autonomy in Quantum Engineering. arXiv:2609.17439 (2026). `https://arxiv.org/abs/2609.17439`

[4] P. Jansen, M.-A. Côté, T. Khot, et al. DiscoveryWorld: A Virtual Environment for Developing and Evaluating Automated Scientific Discovery Agents. arXiv:2406.06769. `https://arxiv.org/abs/2406.06769`

[5] L. Li, W. Chu, J. Langford, X. Wang. Unbiased Offline Evaluation of Contextual-bandit-based News Article Recommendation Algorithms. arXiv:1003.5956. `https://arxiv.org/abs/1003.5956`

[6] S. Cao, Z. Zhang, M. Alghadeer, et al. Agents for self-driving laboratories applied to quantum computing. arXiv:2412.07978. `https://arxiv.org/abs/2412.07978`

[7] C. Shen, W. Zhan, S. Pan, et al. In-situ Self-optimization of Quantum Dot Emission for Lasers by Machine-Learning Assisted Epitaxy. arXiv:2411.00332. `https://arxiv.org/abs/2411.00332`

[8] C. Shen, W. Zhan, J. Tang, et al. Universal Deoxidation of Semiconductor Substrates Assisted by Machine-Learning and Real-Time-Feedback-Control. arXiv:2312.01662. `https://arxiv.org/abs/2312.01662`

[9] N. Muetzel, V. Luu, S. Bey, et al. RHEED pattern classification by a convolutional neural network for the growth of chalcogenide thin films and nanostructures. arXiv:2602.18243. `https://arxiv.org/abs/2602.18243`

A Raw trace excerpts (evidence)

Verbatim excerpts from the raw trace (message index in brackets). They are evidence of what was measured, not instructions to execute.

[151] D1: Fe shutter closed and SeC 540 issued (fese-lab –hardware output)

```
16:26:16
  "ok": true,
    "position_steps": 0,
    "ok": true
  "ok": true,
    "ok": true,
    "sp": 540.0
  "ok": true,
  "sp": 540.0,
  "ok": true,
```

[149] CNN classify log just before D1

```
[16:24:41] 20261012162437674.jpg: STO (2D=0.001 3D=0.089 STO=0.910)
[16:24:51] 20261012162449692.jpg: STO (2D=0.012 3D=0.335 STO=0.653)
[16:25:02] 20261012162501702.jpg: 3D (2D=0.001 3D=0.656 STO=0.343)
[16:25:12] 20261012162507700.jpg: STO (2D=0.001 3D=0.450 STO=0.549)
[16:25:23] 20261012162519706.jpg: 3D (2D=0.011 3D=0.566 STO=0.423)
[16:25:33] 20261012162531723.jpg: 3D (2D=0.021 3D=0.762 STO=0.217)
[16:25:44] 20261012162544042.jpg: 3D (2D=0.028 3D=0.803 STO=0.169)
[16:25:54] 20261012162549644.jpg: 3D (2D=0.035 3D=0.670 STO=0.294)
```

[187] W3D back-fill during the weak-3D plateau (CNN still 2D)

```
[16:41:25] 20261012163944010.jpg: W3D=0.093 D=8.9 spec=185 y_s=321 row_y=251 L=0.05 R=0.04
[16:41:25] 20261012163949592.jpg: W3D=0.099 D=9.7 spec=185 y_s=321 row_y=251 L=0.05 R=0.04
[16:41:25] 20261012163955609.jpg: W3D=0.110 D=10.6 spec=185 y_s=321 row_y=251 L=0.05 R=0.04
[16:41:25] 20261012164001601.jpg: W3D=0.109 D=10.6 spec=186 y_s=322 row_y=251 L=0.06 R=0.04
[16:41:25] 20261012164007612.jpg: W3D=0.095 D=9.2 spec=184 y_s=322 row_y=251 L=0.05 R=0.04
[16:41:25] 20261012164013620.jpg: W3D=0.113 D=10.9 spec=184 y_s=321 row_y=251 L=0.05 R=0.04
[16:41:25] 20261012164019624.jpg: W3D=0.102 D=9.7 spec=183 y_s=322 row_y=251 L=0.05 R=0.05
[16:41:25] 20261012164025617.jpg: W3D=0.108 D=10.5 spec=183 y_s=321 row_y=251 L=0.06 R=0.04
```

[195] CNN at the same time: clean 2D

```
[16:42:25] 20261012164220021.jpg: 2D (2D=0.984 3D=0.016 STO=0.000)
[16:42:35] 20261012164231725.jpg: 2D (2D=0.990 3D=0.010 STO=0.000)
[16:42:46] 20261012164243735.jpg: 2D (2D=0.997 3D=0.003 STO=0.000)
[16:42:56] 20261012164255742.jpg: 2D (2D=0.995 3D=0.005 STO=0.000)
```

[209] D5 stop: both shutters closed; one set-fe returned ok:false (retried)

```
16:57:49 STOP_SEQ
    "position_steps": 0,
    "position_steps": 0,
    "position_steps": 0,
    "position_steps": 0,
16:57:51 SHUTTERS_CLOSED
    "ok": false,
16:57:55 COOLDOWN_STARTED
```

B SOP excerpt used by the operator

The agent receives `/app/docs/SOP.md`: an English gist plus the Chinese original excerpt of `fese_grow_v2.md` (decision tables for strong and weak 3D, stop rule). English gist as given to agents:

Listing:

```
Mapping to the replay CLI: `poll_cnn.py`/`watch_alert.py` -> `fese-lab rheed cnn` / `monitor --channels cnn`; `rheed_w3d.py` ->
      `fese-lab rheed w3d` / `monitor --channels w3d`; `poll_temps.py` -> `fese-lab get-pv` / `monitor --channels temp`.
English gist: strong 3D (CNN p3D>0.5 sustained) -> close Fe, SeC +20 C, wait for 2D recovery (~2 min after the setpoint is
     reached), reopen Fe; if not recovered +20 again. Weak 3D (CNN p3D<0.3 but W3D>0.10 for >=5 frames) -> SeC +5..10 C, keep
     Fe open, re-check W3D; W3D 0.08-0.10 = grey zone, observe. Growth 0.2 layer/min of Fe-open time; stop = close both
     shutters; then return setpoints to standby.
```

C CLI reference given to agents

CLI section of instruction.md (T1/T2)

```
## The `fese-lab` CLI (replay edition)
Run `fese-lab --help` / `fese-lab <cmd> --help`. Every command prints JSON. Each measurement
response carries a **receipt id** (`R0001`, ...) issued by the environment.
```

```
| command | effect | budget |
|---|---|---|
| `fese-lab status` | replay clock, shutters, setpoints, Fe-open minutes, layer estimate (also returns a receipt) | free |
| `fese-lab wait <s>` | advance the replay clock by s seconds (<=900) | free |
| `fese-lab get-pv fe|sec|seb` | latest *measured* temperature (with its measurement time/age) | 1 query |
| `fese-lab rheed cnn|w3d -n K` | latest K analysed RHEED frames: CNN probabilities (2D/3D/STO) or W3D weak-3D index | 1 query
    |
| `fese-lab monitor --interval I --duration S --channels cnn,w3d,temp [alerts]` | polls every I s for up to S s, returns all
    new frames, stops early on an alert. Alerts: `--cnn-3d-alert X --consecutive N`, `--cnn-2d-stable X --consecutive N` (p2D>
    X and p3D<0.1), `--w3d-alert X --w3d-consecutive N`, `--sec-target T --tol t` | 1 query per channel per poll |
| `fese-lab fe open|close`, `fese-lab sec open|close` | shutters (`--motivation "..."`) | action |
| `fese-lab set-sec|set-fe|set-seb <degC> [--rate R] --motivation "..."` | setpoints | action |

Time only advances through `wait`/`monitor`; your thinking time does not consume replay time.
```

D Agent trial action traces

t01-weak3d-dsv4pro: environment audit log (status calls omitted), terminal = unevaluable:sop_valid_unlogged_D4

```
16:36:12  rheed    channel=cnn n=10
          -> 10 records, receipt R0002
16:36:12  rheed    channel=w3d n=10
          -> 10 records, receipt R0003
16:38:12  monitor  interval=12.0 duration=120.0 channels=cnn,w3d cnn_3d_alert=0.5 w3d_ale
          -> duration (receipt R0004)
16:40:42  monitor  interval=30.0 duration=300.0 channels=cnn,w3d cnn_3d_alert=0.5 w3d_ale
          -> W3D>0.1 x5 at 16:40:37 (receipt R0006)
16:40:42  set      device=sec value=565.0
          -> SOP-consistent action, but this setpoint was never run; outcome unevaluable
16:40:42  set      device=sec value=570.0
          -> episode ended: unevaluable:sop_valid_unlogged_D4

agent --motivation strings: weak3D: W3D>0.10 sustained (peak 0.115), CNN p3D<0.05; SeC 560->565 | weak3D: W3D>0.10 sustained (
     peak 0.115), CNN p3D<0.05; SeC 560->570
```

t02-diagnose-dsv4pro: environment audit log (status calls omitted), terminal = None

```
17:19:06  archive  channel=actions from=16:00:00 to=17:20:00
          -> 14 records, receipt R0002
17:19:06  archive  channel=cnn from=16:22:52 to=16:26:30
          -> 20 records, receipt R0003
17:19:06  archive  channel=w3d from=16:36:11 to=16:57:49
          -> 20 records, receipt R0004
17:19:06  archive  channel=w3d from=16:39:33 to=16:57:49
          -> 20 records, receipt R0005
17:19:06  archive  channel=w3d from=16:41:32 to=16:57:49
          -> 20 records, receipt R0006
17:19:06  archive  channel=cnn from=16:40:13 to=16:41:51
          -> 7 records, receipt R0007
17:19:06  archive  channel=w3d from=16:43:32 to=16:57:49
          -> 20 records, receipt R0008
17:19:06  archive  channel=w3d from=16:45:44 to=16:57:49
          -> 20 records, receipt R0009
17:19:06  archive  channel=w3d from=16:47:44 to=16:57:49
          -> 20 records, receipt R0010
17:19:06  archive  channel=w3d from=16:49:44 to=16:57:49
          -> 20 records, receipt R0011
17:19:06  archive  channel=w3d from=16:51:44 to=16:57:49
          -> 20 records, receipt R0012
17:19:06  archive  channel=w3d from=16:53:44 to=16:57:49
          -> 20 records, receipt R0013
17:19:06  archive  channel=w3d from=16:57:26 to=16:57:49
          -> 3 records, receipt R0014
17:19:06  archive  channel=cnn from=16:26:16 to=16:31:06
          -> 18 records, receipt R0015
17:19:06  archive  channel=cnn from=16:34:00 to=16:36:11
          -> 8 records, receipt R0016
```

t03-fullrun-dsv4pro: environment audit log (status calls omitted), terminal = out_of_support:missed_D2

```
16:22:53  rheed    channel=cnn n=5
          -> 5 records, receipt R0002
16:22:53  rheed    channel=w3d n=5
          -> 5 records, receipt R0003
16:22:53  get-pv   device=fe
          ->
16:22:53  get-pv   device=sec
          ->
16:22:53  get-pv   device=seb
          ->
16:25:53  monitor  interval=30.0 duration=600.0 channels=cnn,w3d cnn_3d_alert=0.5 w3d_ale
          -> CNN p3D>0.5 x3 at 16:25:44 (receipt R0007)
```

```
16:26:16  shutter  device=fe op=close
          -> applied at logged action time 16:26:16 (agent issued at 16:25:53, delta +23 s)
16:26:16  set      device=sec value=540.0
          -> applied at logged action time 16:26:16 (agent issued at 16:26:16, delta +0 s)
16:27:56  monitor  interval=20.0 duration=600.0 channels=cnn,w3d,temp cnn_2d_stable=0.8 c
          -> SeC PV 539.9 within tol of 540.0 (measured 16:27:50) (receipt R0008)
16:31:06  ... 5 rejected calls (nonexistent commands, or commands after termination): monitor, reset, restart, new_episode,
     start
16:31:06  set      device=sec value=560.0
          -> episode ended: out_of_support:missed_D2
16:31:06  ... 35 rejected calls (nonexistent commands, or commands after termination): archive, rheed, get-pv, wait, monitor,
     '', reset, restart, init, begin, new, clear, reset_episode, end, abort, recover, rewind, replay, load, config,

agent --motivation strings: 3D预警 | 强3D: CNN p3D>0.5 持续, SeC 520+20 | test
```

t04-weak3d-dsv4pro-r2: environment audit log (status calls omitted), terminal = unevaluable:sop_valid_unlogged_D4

```
16:36:12  rheed    channel=cnn n=5
          -> 5 records, receipt R0002
16:36:12  rheed    channel=w3d n=8
          -> 8 records, receipt R0003
16:39:12  monitor  interval=15.0 duration=180.0 channels=cnn,w3d cnn_3d_alert=0.5 w3d_ale
          -> duration (receipt R0004)
16:40:48  monitor  interval=12.0 duration=150.0 channels=cnn,w3d cnn_3d_alert=0.5 w3d_ale
          -> W3D>0.1 x5 at 16:40:37 (receipt R0006)
16:40:48  get-pv   device=sec
          ->
16:40:48  set      device=sec value=565.0
          -> SOP-consistent action, but this setpoint was never run; outcome unevaluable
16:40:48  ... 2 rejected calls (nonexistent commands, or commands after termination): archive, wait

agent --motivation strings: weak-3D: W3D>0.10 sustained (peak 0.115 at 16:40:37), CNN p3D<0.03 2D-dominant; SeC +5 to 565, keep
      Fe open
```

t05-fullrun-dsv4pro-r2: environment audit log (status calls omitted), terminal = out_of_support:missed_D2

```
16:22:53  rheed    channel=cnn n=5
          -> 5 records, receipt R0002
16:22:53  rheed    channel=w3d n=5
          -> 5 records, receipt R0003
16:22:53  get-pv   device=fe
          ->
16:22:53  get-pv   device=sec
          ->
16:22:53  get-pv   device=seb
          ->
16:25:53  monitor  interval=30.0 duration=300.0 channels=cnn,w3d cnn_3d_alert=0.5 w3d_ale
          -> CNN p3D>0.5 x3 at 16:25:44 (receipt R0007)
16:26:16  shutter  device=fe op=close
          -> applied at logged action time 16:26:16 (agent issued at 16:25:53, delta +23 s)
16:26:16  set      device=sec value=540.0
          -> applied at logged action time 16:26:16 (agent issued at 16:26:16, delta +0 s)
16:27:46  monitor  interval=15.0 duration=300.0 channels=cnn,w3d,temp cnn_2d_stable=0.9 c
          -> SeC PV 538.3 within tol of 540.0 (measured 16:27:41) (receipt R0008)
16:30:16  monitor  interval=15.0 duration=150.0 channels=cnn,w3d cnn_2d_stable=0.9 consec
          -> duration (receipt R0009)
16:31:06  ... 43 rejected calls (nonexistent commands, or commands after termination): monitor, archive, reset, restart, help,
     list, new, new_episode, begin, start, init, replay, reset_episode, load, home, snapshot, rollback, rewind, got
16:31:06  set      device=sec value=560.0
          -> episode ended: out_of_support:missed_D2
16:31:06  shutter  device=fe op=open
          -> episode ended: out_of_support:missed_D2
16:31:06  ... 40 rejected calls (nonexistent commands, or commands after termination): wait, '', archive, get-pv, diagnose,
     mode, switch, control, set-mode, set_mode, mode-switch, enter_diagnose, diagnostic, arch, query, records, data, h

agent --motivation strings: 3D预警 | 3D干预循环: 强3D关Fe后SeC+20 | D2: 2D未恢复继续+20爬坡 | test
```